SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems.* (BL3, BL6)

Activity Tool : Analytical Problem Solving (High)
Assessment Tool Weightage: 40%

QUESTIONS		
Q.No	**Question**	**Bloom's Level**
1	Given a set of relations and sample data, write SQL queries to retrieve specific information using joins and subqueries, and explain the logic used. Bloom's Level: BL3 - Apply	BL4 – Analyze
2	Using relational algebra, formulate expressions to solve complex queries such as retrieving records that satisfy multiple conditions across relations. Bloom's Level: BL4 - Analyse	BL5 – Evaluate
3	Given a real-world scenario (e.g., banking or student system), design a relational schema and construct appropriate SQL queries for data retrieval and updates. Bloom's Level: BL6 – Create	BL6 – Create
4	Analyse a given SQL query with nested subqueries and predict its output, explaining each step of execution. Bloom's Level: BL4 - Analyze	BL6 – Create
5	Compare different SQL approaches (joins vs subqueries) for solving the same problem and evaluate their efficiency. Bloom's Level: BL5 – Evaluate	BL4 – Analyze
6	Design and implement views for a database system to provide restricted data access, and justify their use for security and abstraction. Bloom's Level: BL6 - Create	BL5 – Evaluate

7	Given a database schema, write relational algebra expressions and convert them into equivalent SQL queries, analyzing differences in representation. Bloom's Level: BL4 – Analyze	BL4 – Analyze

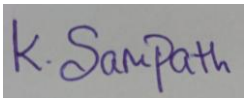
8	Optimize a set of inefficient SQL queries by restructuring them using joins, indexing concepts, or views, and evaluate performance improvements. Bloom's Level: BL5 – Evaluate	BL5 – Evaluate
9	Develop a complex SQL query involving grouping, aggregation, and nested queries to solve a real-world problem. Bloom's Level: BL6 – Create	BL6 – Create
10	Given multiple relations, analyze the query requirements and decide whether to use views, joins, or subqueries, justifying your choice with reasoning. Bloom's Level: BL5 – Evaluate	BL5 – Evaluate

---

Code of Conduct:

I **K. Sampath Chaitanya (192572006)** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understandin g	Clearly interprets problem, identifies all parameters and requirements accurately	Understand s problem with minor gaps	Partial understandin g; misses key elements	Misinterpret s or unable to understand problem
Method Selection	Selects most appropriate method/formul a with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	All calculations accurate;	Minor calculatio n errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretati o n with minor omissions	Limited or unclear interpretation	No interpretatio n

1}

Given a set of relations and sample data, write SQL queries to retrieve specific information using joins and subqueries, and explain the logic used. Bloom's Level: BL3 – Apply

1. CREATE STUDENT TABLE

```
mysql> CREATE TABLE student (
    ->     RegNo INT PRIMARY KEY,
    ->     Name VARCHAR(20),
    ->     Gender CHAR(1),
    ->     DOB DATE,
    ->     MobileNo BIGINT,
    ->     City VARCHAR(20),
    ->     FacNo INT
    -> );
Query OK, 0 rows affected (0.13 sec)

mysql> DESC student;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| RegNo    | int         | NO   | PRI | NULL    |       |
| Name     | varchar(20) | YES  |     | NULL    |       |
| Gender   | char(1)     | YES  |     | NULL    |       |
| DOB      | date        | YES  |     | NULL    |       |
| MobileNo | bigint      | YES  |     | NULL    |       |
| City     | varchar(20) | YES  |     | NULL    |       |
| FacNo    | int         | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
7 rows in set (0.10 sec)
```

2. CREATE DEPARTMENT TABLE4

```
mysql> CREATE TABLE Department (
    ->     DeptNo INT PRIMARY KEY,
    ->     DeptName VARCHAR(20),
    ->     HOD VARCHAR(20)
    -> );
ERROR 1050 (42S01): Table 'department' already exists
mysql> DESC Department;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| DeptNo   | varchar(4)  | NO   | PRI | NULL    |       |
| DeptName | varchar(15) | YES  |     | NULL    |       |
| DeptHead | varchar(4)  | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.06 sec)
```

3. Insert Sample Data

Department

```
mysql> INSERT INTO Department VALUES
    -> (101,'CSE','Mohan'),
    -> (102,'ECE','Raju'),
    -> (103,'EEE','Ramesh');
Query OK, 3 rows affected (0.07 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Department;
+--------+----------+--------+
| DeptNo | DeptName | HOD    |
+--------+----------+--------+
|    101 | CSE      | Mohan  |
|    102 | ECE      | Raju   |
|    103 | EEE      | Ramesh |
+--------+----------+--------+
3 rows in set (0.04 sec)
```

Student

```
mysql> INSERT INTO Student VALUES
    -> (19221151,'Akash','M','2004-01-12',9876543210,'Chennai',101),
    -> (19221152,'Priya','F','2005-02-13',9876543211,'Madurai',102),
    -> (19221153,'Rahul','M','2004-03-14',9876543212,'Salem',101),
    -> (19221154,'Sneha','F','2005-04-15',9876543213,'Coimbatore',103),
    -> (19221155,'Kiran','M','2004-05-16',9876543214,'Trichy',102);
Query OK, 5 rows affected (0.05 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> DESC STUDENT;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| RegNo    | int         | NO   | PRI | NULL    |       |
| Name     | varchar(20) | YES  |     | NULL    |       |
| Gender   | char(1)     | YES  |     | NULL    |       |
| DOB      | date        | YES  |     | NULL    |       |
| MobileNo | bigint      | YES  |     | NULL    |       |
| City     | varchar(20) | YES  |     | NULL    |       |
| FacNo    | int         | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
7 rows in set (0.01 sec)
```

JOIN Queries

1. Display Student Name and Department Name

```
mysql> SELECT Student.Name, Department.DeptName
    -> FROM Student
    -> INNER JOIN Department
    -> ON Student.DeptNo = Department.DeptNo;
+-------+----------+
| Name  | DeptName |
+-------+----------+
| Akash | CSE      |
| Priya | ECE      |
| Rahul | CSE      |
| Sneha | EEE      |
| Kiran | ECE      |
+-------+----------+
5 rows in set (0.00 sec)
```

2}

Using relational algebra, formulate expressions to solve complex queries such as retrieving records that satisfy multiple conditions across relations.Bloom’s Level: BL4 – Analyse

Relations

STUDENT (RegNo, Name, Gender, Dept_No)

DEPARTMENT (DeptNo, DeptName, HOD)

Query 1

Retrieve the names of students who belong to the CSE department.

Relational Algebra:

πName (σDeptName='CSE'(STUDENT ⋈ STUDENT.DeptNo = DEPARTMENT.DeptNo DEPARTMENT))

Query 2

Retrieve the names of female students in the CSE department.

Relational Algebra:

πName(σGender='F' ∧ DeptName='CSE'

(STUDENT ⋈ STUDENT.DeptNo = DEPARTMENT.DeptNo DEPARTMENT))

Query 3

Retrieve the names of students whose HOD is 'Mohan'.

Relational Algebra:

πName(σHOD='Mohan'

(STUDENT ⋈ STUDENT.DeptNo = DEPARTMENT.DeptNo DEPARTMENT))

Query 4

Retrieve the names and department names of all students.

Relational Algebra:

πName,DeptName

(STUDENT ⋈ STUDENT.DeptNo = DEPARTMENT.DeptNo DEPARTMENT)

Query 5

Retrieve the department names having female students.

Relational Algebra:

πDeptName

(σGender='F'

(STUDENT ⋈ STUDENT.DeptNo = DEPARTMENT.DeptNo DEPARTMENT))

Query 6

Retrieve the names of students who belong to the ECE department.

Relational Algebra:

πName

(σDeptName='ECE'

(STUDENT ⋈ STUDENT.DeptNo = DEPARTMENT.DeptNo DEPARTMENT))

Operators Used

Symbol	Meaning
σ	Selection (filters rows based on a condition)
π	Projection (selects required columns)
⋈	Join (combines two relations using a common attribute)
∧	Logical AND (multiple conditions)

Explanation

- Selection (σ) is used to filter records based on one or more conditions.
- Projection (π) is used to retrieve only the required attributes.
- Join (⋈) combines the STUDENT and DEPARTMENT relations using the common attribute DeptNo.

Bottom of Form

3}

relational schema and construct appropriate SQL queries for data retrieval and updates.
Bloom's Level: BL6 – Create

1. Create Student Table

```
mysql> CREATE TABLE student (
    ->     RegNo INT PRIMARY KEY,
    ->     Name VARCHAR(20),
    ->     Gender CHAR(1),
    ->     DOB DATE,
    ->     MobileNo BIGINT,
    ->     City VARCHAR(20),
    ->     FacNo INT
    -> );
Query OK, 0 rows affected (0.13 sec)

mysql> DESC student;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| RegNo    | int         | NO   | PRI | NULL    |       |
| Name     | varchar(20) | YES  |     | NULL    |       |
| Gender   | char(1)     | YES  |     | NULL    |       |
| DOB      | date        | YES  |     | NULL    |       |
| MobileNo | bigint      | YES  |     | NULL    |       |
| City     | varchar(20) | YES  |     | NULL    |       |
| FacNo    | int         | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
7 rows in set (0.10 sec)
```

2. Insert Sample Data

```
mysql> CREATE TABLE faculty
    -> (
    ->     RegNo INT(3),
    ->     Name VARCHAR(15),
    ->     Gender CHAR(1),
    ->     DOB DATE,
    ->     MobileNo BIGINT,
    ->     City VARCHAR(15)
    -> );
ERROR 1050 (42S01): Table 'faculty' already exists
mysql> INSERT INTO faculty
    -> VALUES
    -> (11911511,'mohan','m','2004-01-12',998826973,'mtr'),
    -> (11922112,'raju','m','2005-02-13',908658694,'mgr'),
    -> (11922113,'ramesh','m','2005-03-14',998263548,'mnr'),
    -> (11922154,'ramu','m','2006-04-14',908269279,'mgr'),
    -> (11922115,'san','m','2007-05-17',998269245,'hyt');
Query OK, 5 rows affected (0.06 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> desc faculty;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| RegNo    | int         | YES  |     | NULL    |       |
| Name     | varchar(15) | YES  |     | NULL    |       |
| Gender   | char(1)     | YES  |     | NULL    |       |
| DOB      | date        | YES  |     | NULL    |       |
| MobileNo | bigint      | YES  |     | NULL    |       |
| City     | varchar(15) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

3. Display All Students

```
mysql> SELECT * FROM Student;
+----------+--------+--------+------------+-----------+------------+
| RegNo    | Name   | Gender | DOB        | MobileNo  | City       |
+----------+--------+--------+------------+-----------+------------+
| 19221151 | Mohan  | M      | 2004-01-12 | 990826973 | Madurai    |
| 19221152 | Raju   | M      | 2005-02-13 | 908658694 | Chennai    |
| 19221153 | Ramesh | M      | 2005-03-14 | 998263548 | Trichy     |
| 19221154 | Ram    | M      | 2006-04-14 | 908269279 | Salem      |
| 19221155 | San    | M      | 2007-05-17 | 998269245 | Coimbatore |
+----------+--------+--------+------------+-----------+------------+
5 rows in set (0.28 sec)
```

4. Display students from Chennai.

```
mysql> SELECT *
    -> FROM Student
    -> WHERE City='Chennai';
+----------+------+--------+------------+-----------+---------+
| RegNo    | Name | Gender | DOB        | MobileNo  | City    |
+----------+------+--------+------------+-----------+---------+
| 19221152 | Raju | M      | 2005-02-13 | 908658694 | Chennai |
+----------+------+--------+------------+-----------+---------+
1 row in set (0.06 sec)
```

4}

Analyse a given SQL query with nested subqueries and predict its output, explaining each step of execution.Bloom's Level: BL4 – Analyze

Nested Subquery

```
mysql> SELECT * FROM Department;
+--------+----------+--------+
| DeptNo | DeptName | HOD    |
+--------+----------+--------+
|    101 | CSE      | Mohan  |
|    102 | ECE      | Raju   |
|    103 | EEE      | Ramesh |
+--------+----------+--------+
3 rows in set (0.00 sec)
```

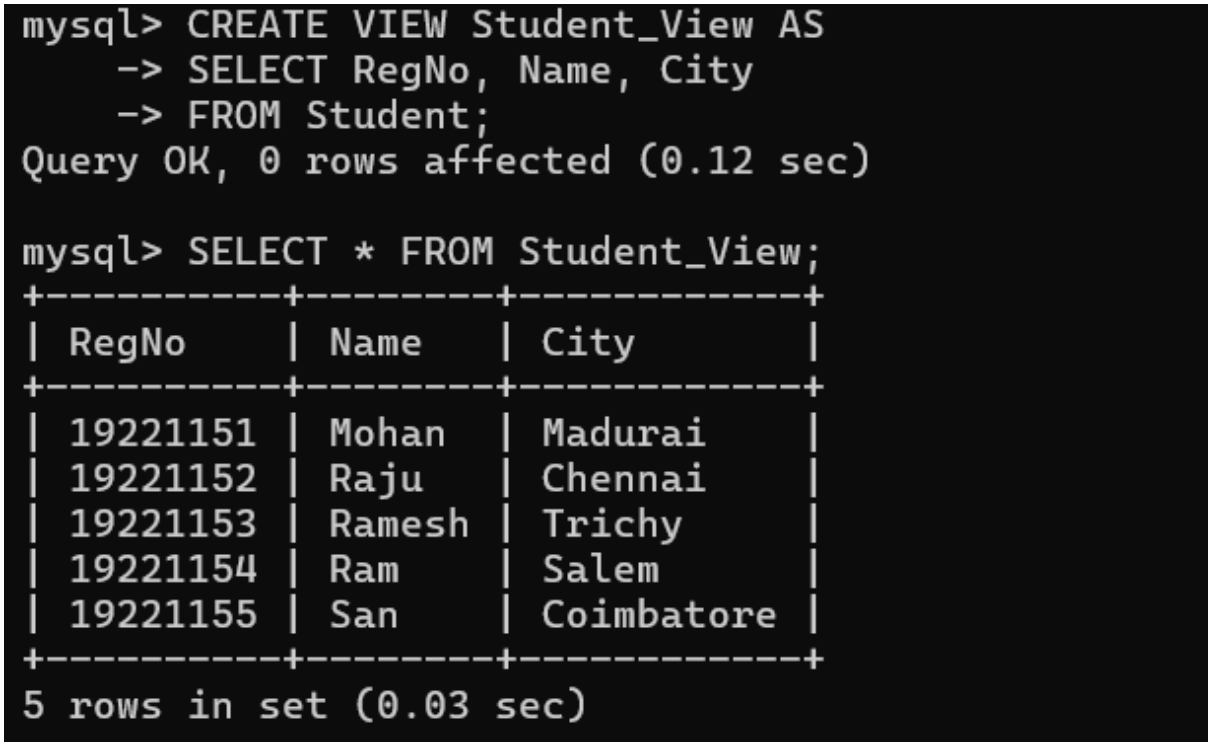
5}

Compare different SQL approaches (joins vs subqueries) for solving the same problem and evaluate their efficiency.Bloom's Level: BL5 – Evaluate

```
mysql> SELECT s.Name, d.DeptName
    -> FROM Student s
    -> INNER JOIN Department d
    -> ON s.DeptNo = d.DeptNo;
+--------+----------+
| Name   | DeptName |
+--------+----------+
| Mohan  | CSE      |
| Raju   | ECE      |
| Ramesh | CSE      |
| Ram    | EEE      |
| San    | ECE      |
+--------+----------+
5 rows in set (0.05 sec)
```

6}

Design and implement views for a database system to provide restricted data access, and justify their use for security and abstraction.Bloom’s Level: BL6 – Create



```
mysql> CREATE VIEW Student_View AS
    -> SELECT RegNo, Name, City
    -> FROM Student;
Query OK, 0 rows affected (0.12 sec)

mysql> SELECT * FROM Student_View;
+----------+--------+------------+
| RegNo    | Name   | City       |
+----------+--------+------------+
| 19221151 | Mohan  | Madurai    |
| 19221152 | Raju   | Chennai    |
| 19221153 | Ramesh | Trichy     |
| 19221154 | Ram    | Salem      |
| 19221155 | San    | Coimbatore |
+----------+--------+------------+
5 rows in set (0.03 sec)
```

7}

Given a database schema, write relational algebra expressions and convert them into equivalent SQL queries, analyzing differences in representation.Bloom’s Level: BL4 – Analyze

Given Database Schema

Student Student_ID Student_Name Dept_ID Marks

Department Dept_ID

Dept_Name

Example 1: Students in the CSE Department

Relational Algebra

πStudent_Name,Dept_Name(σDept_Name=′CSE′(Student$\bowtie$Department

Equivalent SQL

SQL

```
SELECT Student.Student_Name, Department.Dept_Name
    FROM Student
    JOIN Department
    ON Student.Dept_ID = Department.Dept_ID
    WHERE Department.Dept_Name = 'CSE';
```

Analysis

Relational Algebra: Uses σ (Selection), ⋈ (Join), and π (Projection).

SQL: Uses WHERE, JOIN, and SELECT.

Both return the same result, but SQL is more user-friendly and executable.

Example 2: Students Scoring More Than 80

Relational Algebra

πStudent_Name,Marks(σMarks>80(Student))

Equivalent SQL

Example 3: Average Marks Department-wise

Relational Algebra (Extended)

γDept_ID, AVG(Marks)→Avg_Marks(Student)

Equivalent SQL :

```
SELECT Dept_ID,
AVG(Marks) AS Avg_Marks
FROM Student
GROUP BY Dept_ID;
```

Analysis

- Extended RA uses the γ (Grouping) operator.

- SQL uses GROUP BY with aggregate function

SQL is easier to read and directly supported by DBMSs.

Example 4: Student Names with Department Names

Relational Algebra

πStudent_Name,Dept_Name(Student$\bowtie$Department)

Equivalent SQL:

```
SELECT Student.Student_Name,
Department.Dept_Name
FROM Student
JOIN Department
ON Student.Dept_ID = Department.Dept_ID;
```

Analysis

- RA uses the join operator $\bowtie$.
- SQL uses the JOIN ... ON clause.
- Both produce the same output.

Comparison: Relational Algebra vs SQL

Relational Algebra	SQL
Procedural query language	Declarative query language
Uses symbols like σ, π, $\bowtie$	Uses keywords like SELECT, FROM, WHERE
Describes the sequence of operations	Describes the required result
Mainly used for query design and optimization	Used to interact with real databases
Theoretical foundation	Practical implementation
Database Schema	STUDENT(RegNo, Name, Gender, DOB, MobileNo, City)

Query 1

Display the names of students from Chennai.

Relational Algebra

πName(σCity='Chennai'(STUDENT))

```
mysql> SELECT Name
    -> FROM Student
    -> WHERE City='Chennai';
+------+
| Name |
+------+
| Raju |
+------+
1 row in set (0.01 sec)
```

Query 2

Display the registration number and name of male students.

Relational Algebra

πRegNo,Name(σGender='M'(STUDENT))

```
mysql> SELECT RegNo, Name
    -> FROM Student
    -> WHERE Gender='M';
+----------+--------+
| RegNo    | Name   |
+----------+--------+
| 19221151 | Mohan  |
| 19221152 | Raju   |
| 19221153 | Ramesh |
| 19221154 | Ram    |
| 19221155 | San    |
+----------+--------+
5 rows in set (0.00 sec)
```

Query 3

Display all students born after 01-01-2005.

Relational Algebra

σDOB>'2005-01-01'(STUDENT)

```
mysql> SELECT *
    -> FROM Student
    -> WHERE DOB > '2005-01-01';
+----------+--------+--------+------------+-----------+------------+--------+
| RegNo    | Name   | Gender | DOB        | MobileNo  | City       | DeptNo |
+----------+--------+--------+------------+-----------+------------+--------+
| 19221152 | Raju   | M      | 2005-02-13 | 908658694 | Chennai    |    102 |
| 19221153 | Ramesh | M      | 2005-03-14 | 998263548 | Trichy     |    101 |
| 19221154 | Ram    | M      | 2006-04-14 | 908269279 | Salem      |    103 |
| 19221155 | San    | M      | 2007-05-17 | 998269245 | Coimbatore |    102 |
+----------+--------+--------+------------+-----------+------------+--------+
4 rows in set (0.06 sec)
```

8}

Optimize a set of inefficient SQL queries by restructuring them using joins, indexing concepts, or views, and evaluate performance improvements. Bloom's Level: BL5 – Evaluate

Query 1: Inefficient Query (Using Subquery)

Problem: Display the names of students who belong to the CSE department.

```
mysql> SELECT Name
    -> FROM Student
    -> WHERE DeptNo =
    -> (
    -> SELECT DeptNo
    -> FROM Department
    -> WHERE DeptName = 'CSE'
    -> );
+--------+
| Name   |
+--------+
| Mohan  |
| Ramesh |
+--------+
2 rows in set (0.06 sec)
```

Query 2: Optimization Using View

Create a view to simplify repeated queries.

```
mysql> CREATE VIEW Student_Department AS
    -> SELECT s.RegNo,
    ->        s.Name,
    ->        d.DeptName
    -> FROM Student s
    -> INNER JOIN Department d
    -> ON s.DeptNo = d.DeptNo;
Query OK, 0 rows affected (0.03 sec)

mysql> SELECT * FROM Student_Department;
+----------+--------+----------+
| RegNo    | Name   | DeptName |
+----------+--------+----------+
| 19221151 | Mohan  | CSE      |
| 19221153 | Ramesh | CSE      |
| 19221152 | Raju   | ECE      |
| 19221155 | San    | ECE      |
| 19221154 | Ram    | EEE      |
+----------+--------+----------+
5 rows in set (0.06 sec)
```

9}

Develop a complex SQL query involving grouping, aggregation, and nested queries to solve a real-world problem

Scenario

A Student Management System stores student records. The administrator wants to know the number of students in each city and identify cities that have more than one student.

Query 1: Grouping and Aggregation

Display the number of students in each city.

```
mysql> SELECT City, COUNT(*) AS Total_Students
    -> FROM Student
    -> GROUP BY City;
+------------+----------------+
| City       | Total_Students |
+------------+----------------+
| Madurai    |              1 |
| Chennai    |              1 |
| Trichy     |              1 |
| Salem      |              1 |
| Coimbatore |              1 |
+------------+----------------+
5 rows in set (0.06 sec)
```

10}

Given multiple relations, analyze the query requirements and decide whether to use Views, JOINs, or Subqueries, justifying your choice with reasoning.

Bloom's Level: BL5 – Evaluate

Relations

STUDENT(RegNo, Name, Gender, DeptNo, City)

DEPARTMENT(DeptNo, DeptName, HOD)

1. Using JOIN Query

Display the student name along with the department name.

```
mysql> SELECT s.RegNo, s.Name, d.DeptName
    -> FROM Student s
    -> INNER JOIN Department d
    -> ON s.DeptNo = d.DeptNo;
+----------+--------+----------+
| RegNo    | Name   | DeptName |
+----------+--------+----------+
| 19221151 | Mohan  | CSE      |
| 19221153 | Ramesh | CSE      |
| 19221152 | Raju   | ECE      |
| 19221155 | San    | ECE      |
| 19221154 | Ram    | EEE      |
+----------+--------+----------+
5 rows in set (0.00 sec)
```

Reason

- JOIN is used because data is required from both Student and Department tables.
- It is faster and more efficient for retrieving related data from multiple tables.

2. Using SUBQUERY

Query
Display the names of students who belong to the CSE department.

```
mysql> SELECT Name
    -> FROM Student
    -> WHERE DeptNo =
    -> (
    -> SELECT DeptNo
    -> FROM Department
    -> WHERE DeptName = 'CSE'
    -> );
+--------+
| Name   |
+--------+
| Mohan  |
| Ramesh |
+--------+
2 rows in set (0.05 sec)
```